Multi-Source Candidate Data Transformer

Stage 1 Technical Design Document

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1. PIPELINE & SYSTEM WORKFLOW (ARCHITECTURE BREAKDOWN)

The system enforces a strict **unidirectional data flow** designed around the **Ports and Adapters** pattern to isolate I/O from core processing: **(1) Ingest**: Input adapters (CSV, JSON, PDF, DOCX, TXT) read raw files/streams and output a standard raw envelope. **(2) Extract/Parse**: Extracts raw semi-structured text. **(3) Normalize**: Standardizes values (phones, dates, names) into clean formats. **(4) Resolve Identity**: Deterministically matches candidates against existing profiles using indexed hashes of primary keys (emails, phones). **(5) Merge & Conflict Resolution**: Merges records field-by-field, resolving schema collisions using configured rules while building a central **provenance log**. **(6) Validate**: Business validation runs on the merged golden record (Pydantic v2). **(7) Project**: Configurable mapper reshapes output JSON dynamically.

2. CANONICAL DATA SCHEMA & NORMALIZATION RULES

The internal **Canonical Candidate Record** defines a single point of truth. All attributes are normalized upon entry:

- **Names**: Stripped of honorifics (Mr., Dr.), parsed into Title Case. Concat into *full_name*.
- **Phone Numbers**: Parsed, stripped of symbols, and formatted strictly to **E.164** (e.g., +15551234567). Default country applied if missing.
- **Emails**: Trimmed, validated against RFC 5322, lowercased to ensure deterministic matching.
- **Locations**: Structured as 3-tuple *[city, region, country]*. Country codes mapped to **ISO-3166 alpha-2** (uppercase).
- **Dates**: Standardized to **YYYY-MM** or **YYYY-MM-DD**. 'Present' / 'Current' maps to null to represent active status.
- **Skills**: Flat strings mapped to standard taxonomy terms (e.g. 'js', 'javascript' -> 'JavaScript') to ensure clean search indexing.

3. IDENTITY RESOLUTION, MERGE POLICY & CONFIDENCE ASSIGNMENT

Identity Resolution (Matching): Records are merged if they share any normalized email or phone number. A deterministic matcher computes cryptographic hashes of emails/phones for instant lookup.
Conflict Resolution (Field-Level Survivorship): We use a configurable, attribute-level priority cascade: **(1) Source Priority**: Admin-defined trust hierarchy (e.g., *recruiter_notes* [1.0] > *ats_json* [0.8] > *resume_pdf* [0.6]). **(2) Extraction Confidence**: Parsers supply a confidence rating. If source priority ties, higher confidence wins. **(3) Source Recency**: Later source update timestamps break ties.
Collection Merging: Flat lists (like skills) are union-deduplicated. Hierarchical lists (work/education) are matched by key (e.g., company+title) and merged internally; otherwise, they are appended. **Provenance** is tracked per-field: *provenance: [{ field, source, method }]*. **Overall Confidence** is calculated as the weighted average of surviving attribute extraction confidences.

4. RUNTIME CONFIGURABLE PROJECTION ENGINE

To satisfy downstream consumer schemas without modifying code, we implement a runtime JSON projection engine. The engine reads a configuration specifying: **(1) Subset selection**: Filters fields. **(2) Remapping (from)**: Paths like *emails[0]* mapped to a new key *primary_email*. **(3) Custom normalizations**: Formats fields at output time (e.g. anonymizes skills, masks names, forces E.164). **(4) Missing value handling**: Configures fallback actions (*null*, *omit* from JSON, or raise a fatal *error*). This maintains clean separation between canonical storage and API display layers.

5. EDGE CASES, ERROR RECOVERY & STRATEGIC DESCOPING

- **Garbage/Empty Sources**: If an adapter receives malformed/empty files, it raises an *IngestionWarning*, generates an empty record wrapper, and allows the remaining sources to run without crashing the batch.
- **Conflicting Names (e.g. Jon Doe vs Johnathan Doe)**: Identity resolver matches them via email. The Merge Engine retains 'Johnathan Doe' if it came from the higher-priority ATS JSON source, but records 'Jon Doe' in the provenance file notes.
- **Overlapping Work Timelines**: If merged dates result in overlapping jobs, a validation warning is logged, but records are preserved to prevent loss of candidate experience data.
- **Deliberate Descope (Time Pressure)**: We descoped: (1) Fuzzy matching names/companies via NLP/LLM (leveraging strict identifier matching instead), (2) Real-time external API calls for skill taxonomy verification (using an in-memory taxonomy dictionary instead), and (3) A database layer (saving output directly to JSON files).